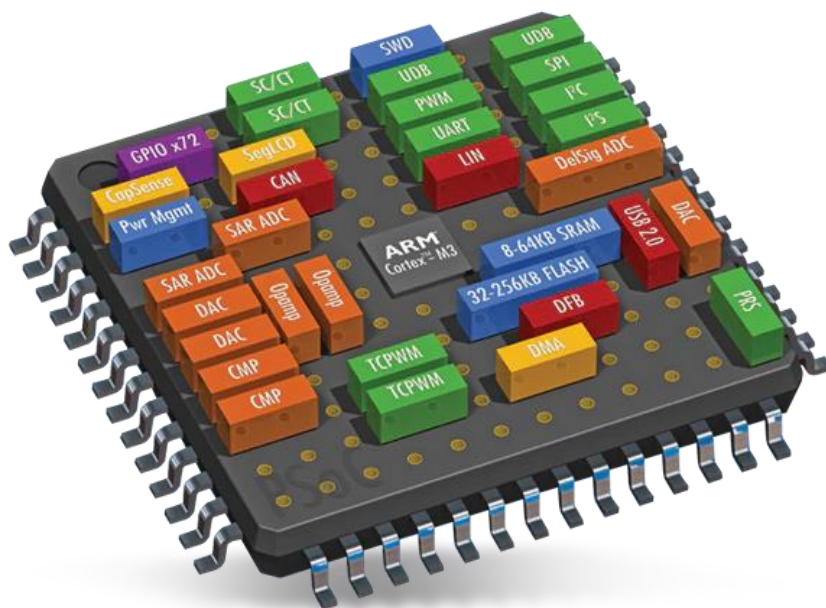




Model Curriculum

Chip Design Associate (O-Level 'Chip Design')

QP Code: NIE/ELE/Q0101

Version: 1.0

NSQF Level: 4

Training Parameters

Sector	Electronics
Sub-Sector	Integrated Electronics and Circuits
Occupation	VLSI Design
Country	India
NSQF Level	4
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 2152.0200
Minimum Educational Qualification and Experience	Completed 12th or equivalent in Science with Physics and Maths or Completed 2nd Year of 3 Years Diploma in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches after class 10th or Acquired NSQF Level 3.5 in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches with 1.5 years of Experience or Acquired NSQF Level 3 in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches with 3-year relevant experience
Pre-Requisite License or Training	-
Minimum Job Entry Age	18 Years
Last Reviewed On	30/5/2024
Next Review Date	30/5/2027
NSQC Approval Date	30/5/2024
QP Version	1.0
Model Curriculum Creation Date	30/5/2024
Model Curriculum Valid Up to Date	30/5/2027
Model Curriculum Version	1.0
Duration of the Course	450 Hours

Program Overview

This section summarises the end objectives of the program along with its duration

Training Outcomes

- Comprehensive understanding of VLSI fundamentals, CMOS transistor theory, and logic design principles.
- Proficiency in Verilog RTL coding, simulation, and verification for digital circuit design.
- Ability to perform static timing analysis, develop timing constraints, and optimize designs for timing closure.
- Knowledge of FPGA architecture, programming, synthesis, optimization, and debugging techniques.
- Hands-on experience through practical labs and projects, preparing students for entry-level roles in the semiconductor industry or FPGA design.

Compulsory Modules

The table lists the modules and their duration corresponding to the Compulsory NOS of the Qualification file.

NOS/Module Name	Training Duration (Hours)		
	Th.	Pr.	Total
NOS1: VLSI Fundamentals NOS Code: NIE/ELE/N0101	25	35	60
NOS2: II. Verilog RTL coding for Synthesis NOS Code: NIE/ELE/N0102	25	35	60
NOS3: Static Timing Analysis of VLSI Circuits NOS Code: NIE/ELE/N0103	25	35	60
NOS4: FPGA Architecture and Programming NOS Code: NIE/ELE/N0104	25	35	60
NOS5: Employability Skills NOS Code : DGT/VSQ/N0102	0	0	60
NOS6: OJT/Project NOS Code : NIE/ELE/N0112	0	0	150
Duration (in Hours)	100	140	450

Module Details

Module 1(NOS1): VLSI Design Fundamentals

NOS Code: NIE/ELE/N0101

Terminal Outcomes:

After completion of the module, the students shall be able to:

- To get overview of VLSI Design cycle and fundamentals of CMOS Transistor Theory
- To get knowledge of CMOS Inverter Characteristics and Logic Design
- Understanding the Transistor schematic, Layout and on-chip wire modelling
- To get overview of Delay and Timing analysis.

Detailed Syllabus

1. Introduction to VLSI Design
 - Overview of VLSI technology and its significance
 - Introduction to VLSI design flow: Front-end and Back-end
2. CMOS Transistor Theory
 - Fundamentals of MOSFET operation
 - MOSFET characteristics and modeling
 - MOSFET scaling trends and technology advancements
3. CMOS Inverter Characteristics
 - CMOS inverter operation and voltage transfer characteristics
 - Noise margin analysis and power consumption considerations
 - Inverter sizing and optimization techniques
4. CMOS Logic Design
 - Introduction to basic logic gates: AND, OR, NAND, NOR
 - Combinational logic design using CMOS technology
 - Designing complex logic functions using CMOS logic gates
5. Transistor Level Schematics and Layouts
 - Transistor level design techniques
 - Schematic design and layout considerations
 - Design rules, metal layers, and interconnect routing
6. On-Chip Wire Modeling
 - Introduction to on-chip interconnects
 - Wire parasitics and their impact on circuit performance
 - Modeling and simulation of on-chip wires
7. Bonding Diagram, Packaging, and Assembly
 - Overview of packaging technologies and assembly processes
 - Bonding diagram design and considerations
 - Packaging trends and challenges
8. Gate Delays and Logical Effort
 - Gate delay modeling and estimation
 - Introduction to logical effort analysis
 - Determining the best delay-power trade-off for logic gates using P/N ratio
9. Combinational Logic Circuit Critical Path Optimization
 - Identifying critical paths in combinational circuits
 - Gate sizing and logic restructuring techniques

- Timing optimization for improved performance
10. Timing in Sequential Circuits
- Introduction to sequential circuits: flip-flops, registers
 - Setup and hold time analysis
 - Sequential circuit timing considerations and optimization techniques

Key Learning Outcomes:

Duration: 25 hours	Duration: 35 hours
Theory	Practical
- Overview of VLSI technology and its significance. - Able to understand the Introduction to VLSI design flow: Front-end and Back-end	- Apply knowledge of MOSFET characteristics to analyse and simulate transistor behaviour. - Use simulation tools to explore MOSFET scaling trends and technology advancements.
- Able to understand CMOS inverter operation and voltage transfer characteristics - Evaluate noise margin and power consumption in CMOS inverters.	- Implement CMOS inverter circuits and analyse their voltage transfer characteristics. - Optimize inverter sizing and evaluate noise margins and power consumption through practical experimentation.
- Implement transistor level design techniques in schematic and layout design. - Design bonding diagrams considering packaging trends and challenges.	- Design and simulate complex logic functions using CMOS logic gates, focusing on performance and power considerations.
- Able to understand the Gate delay modelling and estimation, Introduction to logical effort analysis - Determining the best delay-power trade-off for logic gates using P/N ratio	- Create transistor-level schematics and layouts adhering to design rules and interconnect routing guidelines. - Evaluate the impact of layout choices on circuit performance through practical layout simulations.
Classroom Aid	
Charts, Models, Video presentations, Simulation equipment, Whiteboard/Smart Board, Headphones with mic, Marker	
Tools, Equipment, and Other Requirements	
Vivado Design Suit, Modelsim	

Module 2(NOS2): Verilog RTL coding for Synthesis

NOS Code: NIE/ELE/N0102

Terminal Outcomes:

After successful completion of the module, the students shall be able to:

- Understand the Design Cycle of VLSI.
- Understand Verilog programming syntax, Level of Abstraction in Verilog programming, and test bench simulation.
- Design and Develop IPs for VLSI using Verilog
- Emulate, debug & Characterize reusable IPs

Detailed Syllabus

1. Introduction to VLSI Design (2 hours)
 - Overview of VLSI technology and its applications
 - Introduction to the VLSI design flow
2. RTL Design Methodology (4 hours)
 - Basics of Register Transfer Level (RTL) design
 - Overview of RTL design process and methodology
 - Introduction to Hardware Description Languages (HDLs) such as Verilog or VHDL
3. Digital Logic Design Principles (4 hours)
 - Combinational and sequential logic design concepts
 - Boolean algebra and logic gates
 - Flip-flops, registers, and state machines
4. RTL Design Using HDL (6 hours)
 - Introduction to Verilog syntax and constructs
 - Designing combinational and sequential logic using HDL
 - Writing RTL code for basic digital circuits
5. RTL Simulation and Verification (6 hours)
 - Functional verification techniques for RTL designs
 - Introduction to test benches and testbench development
 - Simulation and debugging of RTL designs
6. Timing Constraints and Analysis (4 hours)
 - Introduction to timing constraints in RTL design
 - Timing analysis and optimization techniques
 - Setup and hold time violations and resolution

Practical Component :

1. RTL Design and Implementation Labs (40 hours)
 - Hands-on lab exercises to reinforce RTL design concepts
 - Designing and implementing digital circuits using HDL
 - Simulating and verifying the functionality of RTL designs
2. Design Synthesis and Optimization Labs (15 hours)
 - Introduction to synthesis tools and libraries
 - Synthesizing RTL designs for target technologies
 - Optimizing design for area, power, and performance
3. Timing Analysis and Closure Labs (10 hours)
 - Timing analysis using industry-standard tools
 - Constraint development and application
 - Timing closure techniques to meet design requirements

Key Learning Outcomes:

Duration: 25 hours	Duration: 35 hours
Theory	Practical
Explain the significance of VLSI technology and its applications in modern electronic systems.	Writing RTL code for basic digital circuits
-Differentiate between combinational and sequential logic design approaches. -Apply Boolean algebra and logic gates to design basic digital circuits, including flip-flops, registers, and state machines.	Validate RTL designs through simulation using test benches and debugging techniques.
-Develop RTL code for both combinational and sequential logic circuits using Verilog HDL -Create and implement test benches for RTL simulations, and analyze simulation results for debugging.	- Apply functional verification methods to validate RTL designs' functionality. - Develop and execute test benches for comprehensive simulation and debugging of Verilog RTL code.
-Define timing constraints and their importance in ensuring proper circuit operation. -Perform timing analysis, identify setup and hold time violations, and apply optimization techniques to meet timing requirements.	- Define and apply timing constraints in RTL design to meet timing requirements. - Analyze and resolve setup and hold time violations through timing analysis and optimization strategies.
Classroom Aid	
Video presentations, Laboratory equipment, Whiteboard/Smart Board, Marker, Headphones with mic.	
Tools, Equipment, and Other Requirements	
Vivado Design Suit, Modelsim	

Module 3(NOS3): Static Timing Analysis of VLSI Circuits

NOS Code: NIE/ELE/N0103

Terminal Outcomes:

After completion of the module, the students shall be able to:

- To Understand the Static Timing Analysis
- To understand the ECO fixes and timing closure for the VLSI Circuits

Detailed Syllabus

Overview of VLSI STA

Introduction to Timing Analysis, Combinational circuit timing- races and hazards, sequential circuit timing-set up and hold timing, maximum frequency of operation, Practical Examples of Setup and Hold time Violations and its solution. Timing constraints for synthesis. Circuit synthesis and timing analysis.

Timing performance

Techniques to improve timing performance, including pipeline, retiming etc. clock skew, open loop and closed loop timing. Block level and chip level timing analysis.

.STA using EDA tools

Static timing analysis using EDA tools. Timing analysis after synthesis and place and route, Timing report analysis. ECO flow and sign off checks. Timing closure.

Key Learning Outcomes:

Duration: 25 hours	Practical: 35 hours
Theory	Practical
- Describe the key components of STA, including timing paths, constraints, and timing reports. - Analyze timing issues in combinational circuits, including races and hazards.	- Conduct static timing analysis (STA) using industry-standard tools to analyze timing paths in VLSI circuits. - Identify and resolve timing issues such as setup and hold time violations through practical exercises.
- Discuss how timing constraints influence the synthesis process and timing optimization. - Evaluate timing performance metrics such as maximum frequency of operation and clock skew.	- Implement timing constraints in EDA tools and analyze their impact on circuit timing during synthesis. - Evaluate the effects of clock skew and implement open loop/closed loop timing adjustments for optimization.
- Perform timing analysis at both block level and chip level to ensure overall system timing integrity.	- Utilize EDA tools for static timing analysis post-synthesis and place-and-route stages.

- Understand the differences and challenges involved in block-level versus chip-level timing analysis.	- Execute Engineering Change Order (ECO) flows to address design modifications while ensuring timing closure.
Classroom aids	
Video presentations, Laboratory equipment, Whiteboard/Smart Board, Marker, Headphones with mic.	
Tools, Equipment and Other Requirements	
ZynQ7000 SoC Evaluation board, Vivado Software, JTAG Cable	

Module 4(NOS4): FPGA Architecture and Programming

NOS Code: NIE/ELE/N0104

Terminal Outcomes:

After completion of the module, the students shall be able to:

- To get overview of Field-Programmable Gate Arrays
- To get knowledge of Hardware Description Language
- Understanding the FPGA design techniques, verification and debugging

Detailed Syllabus

Introduction to FPGAs:

- Overview of Field-Programmable Gate Arrays (FPGAs) and their applications
- Advantages of using FPGAs in digital design

FPGA Architecture and Components:

- Understanding the internal architecture of FPGAs
- Basic components of an FPGA: Look-Up Tables (LUTs), flip-flops, interconnects, etc.
- Introduction to FPGA families and resources

1Hardware Description Languages (HDLs)

- Introduction to HDLs: Verilog
- Syntax and data types in HDL
- Writing RTL code for FPGAs

FPGA Design Flow:

- Overview of the FPGA design flow and methodology
- Design entry, synthesis, placement, routing, and bitstream generation
- Introduction to design constraints

FPGA Design Optimization Techniques:

- Timing constraints and analysis in FPGA designs
- Pipelining, retiming, and other optimization techniques
- Trade-offs between area, power, and performance

FPGA Verification and Debugging:

- Introduction to FPGA simulation and verification
- Writing testbenches and test vectors
- Debugging techniques for FPGA designs

Key Learning Outcomes:

Duration: 25 hours	Practical: 35 hours
Theory	Practical
- Overview of Field-Programmable Gate Arrays (FPGAs) applications, Advantages of using FPGAs in digital design - Describe the internal architecture of FPGAs, including key components such as Look-Up Tables (LUTs), flip-flops, and interconnects.	- Utilize FPGA development boards to explore and understand the functionalities of these components through practical experiments.
- Master the syntax and data types in HDLs, specifically focusing on Verilog for FPGA programming. - Write RTL code targeting FPGAs and demonstrate understanding of FPGA-specific coding practices.	- Write and implement RTL code using Verilog HDL for FPGA designs. - Implement design constraints and analyse their effects on FPGA design synthesis and performance.
- Outline the FPGA design flow stages, including design entry, synthesis, placement, routing, and bitstream generation. - Apply timing constraints and conduct timing analysis for FPGA designs to meet performance requirements.	- Apply timing constraints and conduct timing analysis for FPGA designs. - Perform FPGA simulation and verification using test benches and test vectors.
Classroom aids	
Video presentations, Laboratory equipment, Whiteboard/Smart Board, Marker, Headphones with mic.	
Tools, Equipment and Other Requirements	
ZynQ7000 SoC Evaluation board, Vivado Software, JTAG Cable	

Module 5 (NOS5): Employability Skills

NOS Code: DGT/VSQ/N0102

Terminal Outcomes:

After completing this program, participants will be able to:

- Outline the importance of Employability Skills for the current job market and future of work.
- List different learning and employability-related GOI and private portals and their usage.
- Research and prepare a note on different industries, trends, required skills, and the available opportunities.

Key Learning Outcomes

Duration: 60:00

Key Learning Outcomes

Constitutional Values – Citizenship

- Explain constitutional values, including civic rights and duties, citizenship, responsibility towards society, and personal values and ethics such as honesty, integrity, caring, and respecting others that are required to become a responsible citizen.
- Demonstrate how to practice different environmentally sustainable practices.

Becoming a Professional in the 21st Century

- Discuss relevant 21st-century skills required for employment.
- Highlight the importance of practicing 21st-century skills like Self-Awareness, Behavior Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn, etc., in personal or professional life.
- Create a pathway for adopting a continuous learning mindset for personal and professional development.

Basic English Skills

- Show how to use basic English sentences for everyday conversation in different contexts, in person and over the telephone.
- Read and understand text written in basic English.
- Write a short note/paragraph/letter/e-mail using correct basic English.

Career Development & Goal Setting

- Create a career development plan.
- Identify well-defined short- and long-term goals.

Communication Skills

- Demonstrate how to communicate effectively using verbal and nonverbal communication etiquette.
- Write a brief note/paragraph on a familiar topic.
- Explain the importance of communication etiquette, including active listening for effective communication.
- Role-play a situation on how to work collaboratively with others in a team.

Diversity and Inclusion

- Demonstrate how to behave, communicate, and conduct appropriately with all genders and PwD.
- Discuss the significance of escalating sexual harassment issues as per the POSH act.

Financial and Legal Literacy

- Discuss various financial institutions, products, and services.
- Demonstrate how to conduct offline and online financial transactions safely and securely and check passbook/statement.
- Explain the common components of salary such as Basic, PF, Allowances (HRA, TA, DA, etc.), tax deductions.
- Calculate income and expenditure for budgeting.
- Discuss legal rights, laws, and aids.

Essential Digital Skills

- Describe the role of digital technology in day-to-day life and the workplace.
- Demonstrate how to operate digital devices and use the associated applications and features safely and securely.
- Demonstrate how to connect devices securely to the internet using different means.
- Follow the dos and don'ts of cybersecurity to protect against cybercrimes.
- Discuss the significance of displaying responsible online behavior while using various social media platforms.
- Create an email id and follow email etiquette to exchange emails.
- Show how to create documents, spreadsheets, and presentations using appropriate applications.
- Utilize virtual collaboration tools to work effectively.

Entrepreneurship

- Explain the types of entrepreneurship and enterprises.
- Discuss how to identify opportunities for potential business, sources of funding, and associated financial and legal risks with its mitigation plan.
- Describe the 4Ps of Marketing-Product, Price, Place, and Promotion and apply them as per requirement.
- Create a sample business plan for the selected business opportunity.

Customer Service

- Classify different types of customers.
- Demonstrate how to identify customer needs and respond to them in a professional manner.
- Discuss various tools used to collect customer feedback.
- Discuss the significance of maintaining hygiene and dressing appropriately.

Getting ready for Apprenticeship & Jobs

- Draft a professional Curriculum Vitae (CV).
- Use various offline and online job search sources to find and apply for jobs.
- Discuss the significance of maintaining hygiene and dressing appropriately for an interview.
- Role-play a mock interview.
- List the steps for searching and registering for apprenticeship opportunities.

Annexure

Trainer Requirements

Trainer Prerequisites						
Minimum Educational Qualification	Specialization	Relevant Industry Experience		Training/Assessment Experience		Remarks
		Years	Specialization	Years	Specialization	
B.Tech or Equivalent as per NCeF	Electronics/ Electronics & Communication/ Electrical/ Electrical and Electronics/Instrumentation / Electronics & Instrumentation / Instrumentation & Control /Biomedical /Computer Science/Information Technology	2	VLSI Design	-	-	-

Trainer Certification*	
Domain Certification	Platform Certification
Certified to TOT for the job role: NIE/ELE/Q0101 “Chip Design Associate (O-Level ‘Chip Design’)” or equivalent as per NCeF. Minimum accepted score is 80%	Recommended that the Trainer is certified for the Job Role: “Trainer (VET and skills)”, mapped to the Qualification Pack: “MEP/Q2601, v2.0” or equivalent as per NCeF. Minimum accepted score is 80%

Assessor Requirements

Assessor Prerequisites						
Minimum Educational Qualification	Specialization	Relevant Industry Experience		Training/Assessment Experience		Remarks
		Years	Specialization	Years	Specialization	
B.Tech or Equivalent as per NCeF	Electronics/ Electronics & Communication/ Electrical/ Electrical and Electronics/Instrumentation/ Electronics & Instrumentation / Instrumentation & Control /Biomedical /Computer Science/Information Technology	3	VLSI Design	-	-	-

Assessor Certification*	
Domain Certification	Platform Certification
Certified to TOA for the job role: NIE/ELE/Q0101 “Chip Design Associate (O-Level ‘Chip Design’) ” or equivalent as per NCeF. Minimum accepted score is 80%	Recommended that the assessor is certified for the Job Role: “Trainer (VET and skills)”, mapped to the Qualification Pack: “MEP/Q2701, v2.0” or equivalent as per NCeF. Minimum accepted score is 80%

Assessment Strategy

Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the defined learning outcomes and assessment criteria.

The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. Competence acquired by the candidate can be obtained by conducting Theory (Online), Practical assessment, internal assessment, Project/Presentation/ Assignment, Major Project. The emphasis is on the practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is required to pass all OUTCOMES individually based on the passing criteria.

About Examination Pattern:

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns roll number.
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
5. Pass percentage would be 50% marks in each component.
6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
9. The examination will be conducted in English language only.

Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

References

Glossary

Term	Description
Declarative Knowledge	Declarative knowledge refers to facts, concepts and principles that need to be known and/or understood in order to accomplish a task or to solve a problem.
Key Learning Outcome	Key learning outcome is the statement of what a learner needs to know, understand and be able to do in order to achieve the terminal outcomes. A set of key learning outcomes will make up the training outcomes. Training outcome is specified in terms of knowledge, understanding (theory) and Skills (practical application).

OJT (M)	On-the-job training (Mandatory); trainees are mandated to complete specified hours of training on site
OJT (R)	On-the-job training (Recommended); trainees are recommended the specified hours of training on site
Procedural Knowledge	Procedural knowledge addresses how to do something, or how to perform a task. It is the ability to work, or produce a tangible work output by applying Cognitive, affective or psychomotor skills.
Training Outcome	Training outcome is a statement of what a learner will know, understand And be able to do upon the completion of the training.
Terminal Outcome	Terminal outcome is a statement of what a learner will know, understand and be able to do upon the completion of a module. A set of terminal outcomes help to achieve the training outcome.

Acronyms and Abbreviations

Term	Description
QP	Qualification Pack
NSQF	National Skills Qualification Framework
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards